

Solution example

Consider the cascade connection of the following two first-order systems:

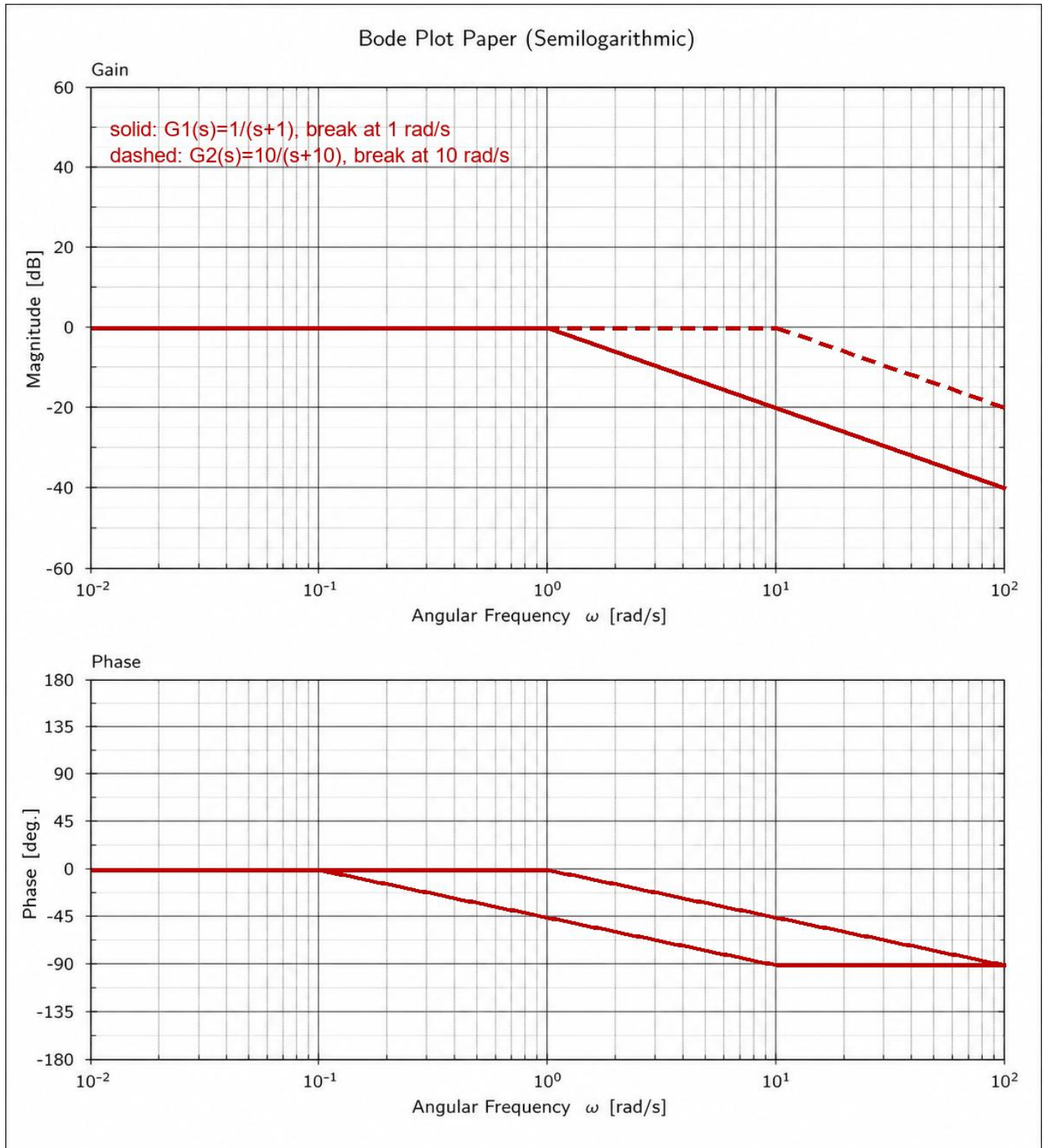
$$G_1(s) = 1/(s+1), \quad G_2(s) = 10/(s+10)$$

The overall transfer function is $G(s) = G_1(s)G_2(s) = 10/[(s+1)(s+10)] = 10/(s^2+11s+10)$.

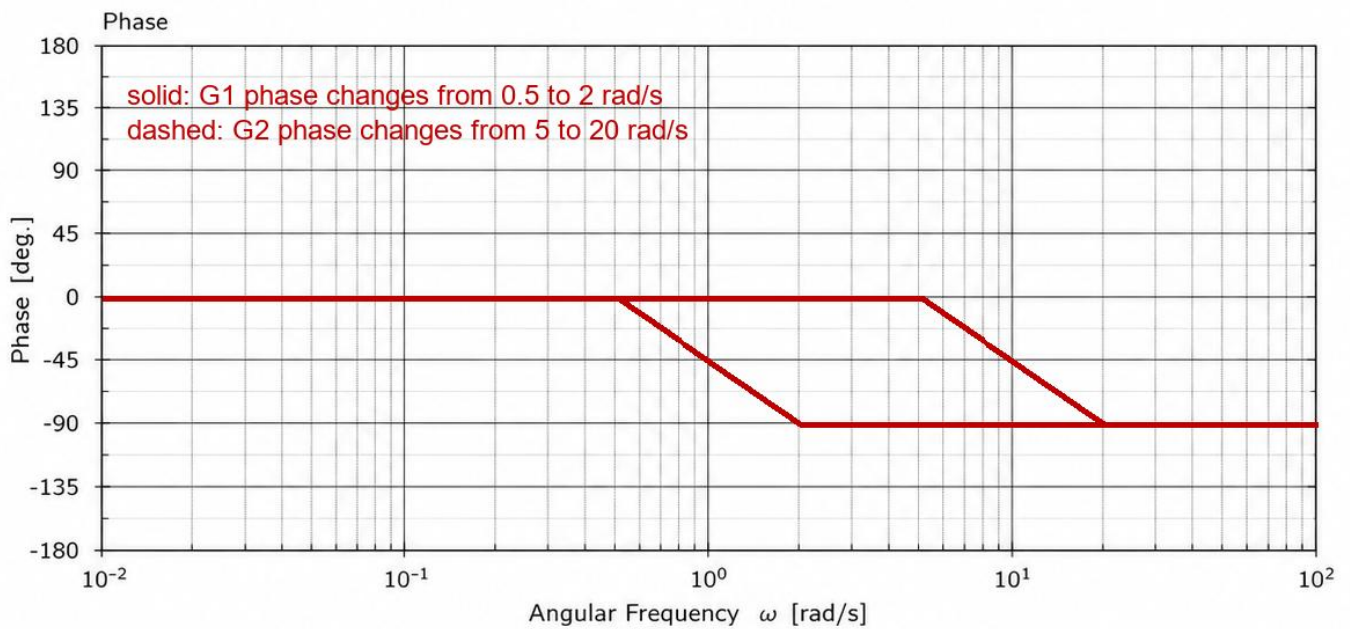
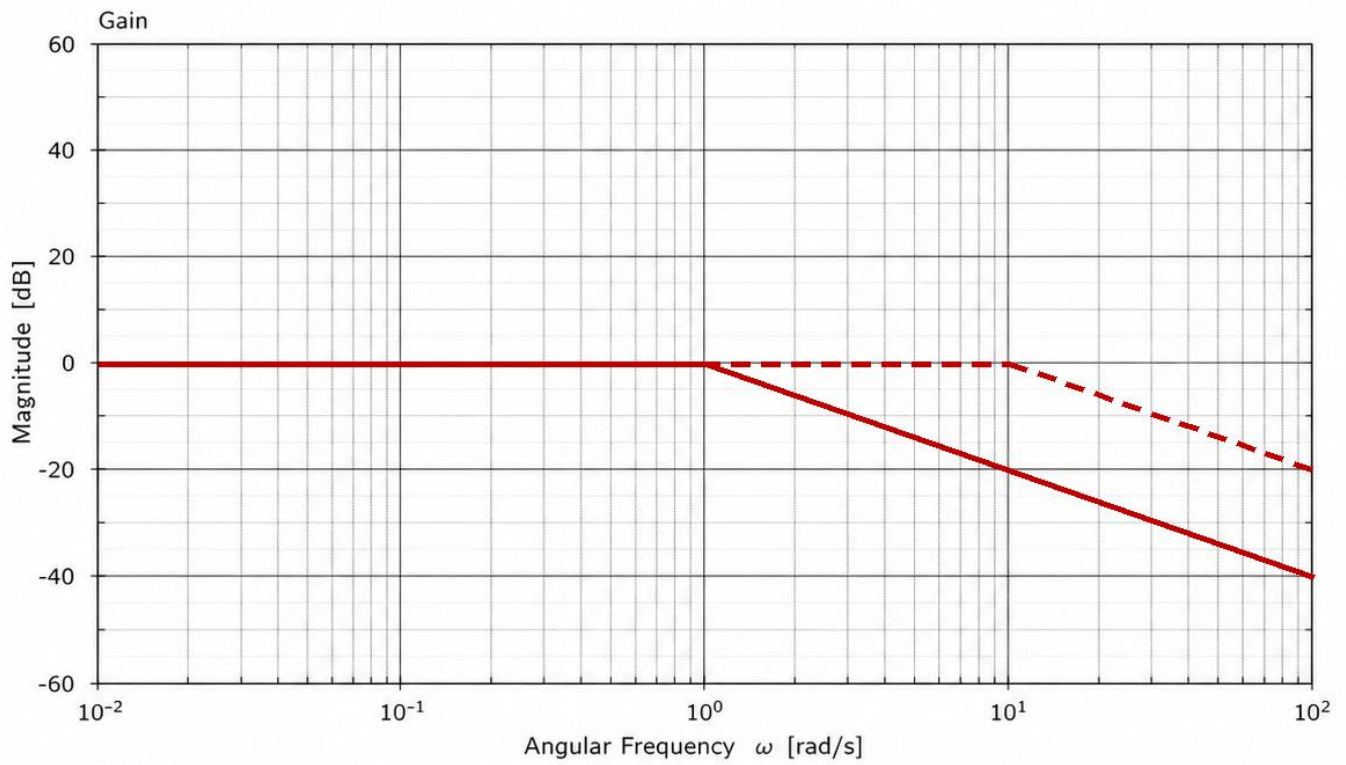
Q1. Bode Plot

(1) Approximate Bode magnitude and phase plots of $G_1(s)$ and $G_2(s)$.

Red solid: $G_1(s)$, red dashed: $G_2(s)$. Breakpoint frequencies are 1 rad/s and 10 rad/s.



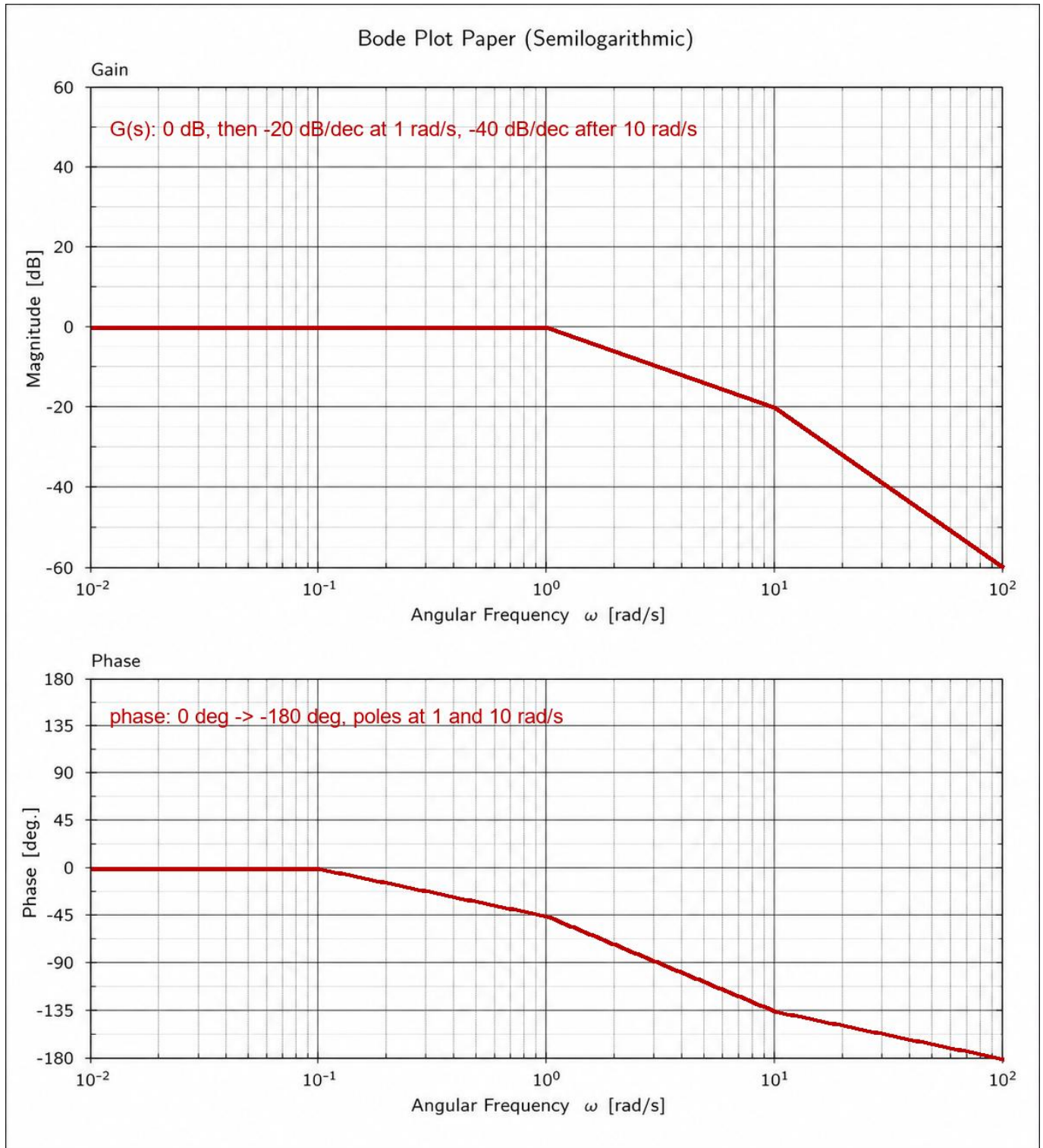
Bode Plot Paper (Semilogarithmic)



Q1. Bode Plot (continued)

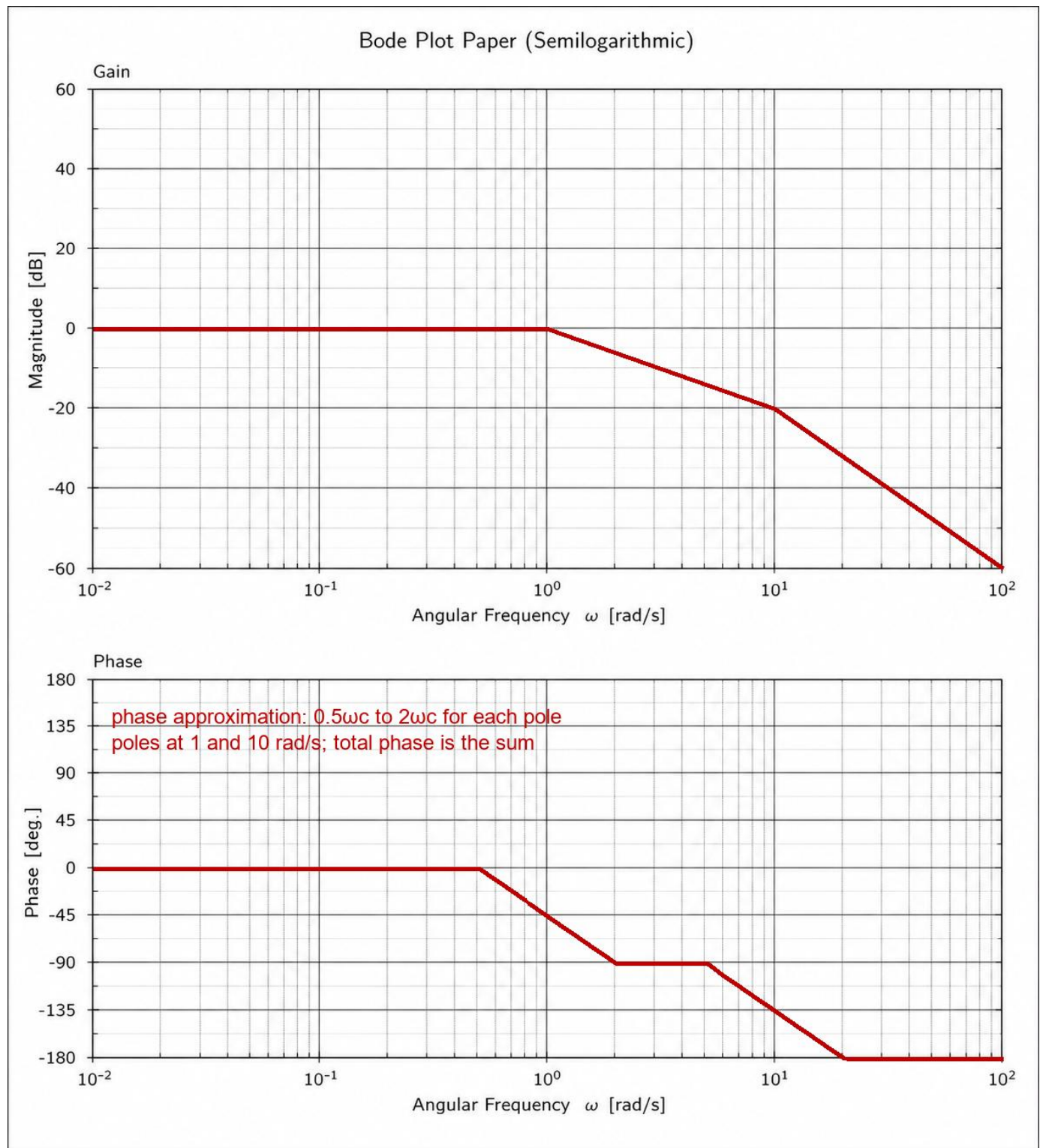
(2) Approximate Bode magnitude and phase plots of the overall transfer function $G(s)$.

The magnitude and phase are obtained by adding the two first-order contributions in dB and degrees.



Q1. Bode Plot (continued)

(2) Approximate Bode magnitude and phase plots of the overall transfer function $G(s)$.



Q2. Response to a Ramp Input

Assume the input is the unit ramp signal $r(t) = t u(t)$, $R(s) = 1/s^2$.

Derive the output response $y(t)$ as a function of time for the overall transfer function $G(s)$.

$$Y(s) = G(s)R(s) = 10/[s^2(s+1)(s+10)].$$

Partial fraction expansion:

$$10/[s^2(s+1)(s+10)] = 1/s^2 - 11/(10s) + 10/[9(s+1)] - 1/[90(s+10)].$$

Taking the inverse Laplace transform gives

$$y(t) = t - 11/10 + (10/9) \exp(-t) - (1/90) \exp(-10t), \quad t \geq 0.$$

Thus the steady ramp response has slope 1 and a constant lag of 11/10.

Q3. Transfer Functions of the Component Systems

Write the transfer functions of the two first-order systems separately. Also identify the time constant and DC gain of each system.

$$G_1(s) = 1/(s+1) = 1/(1*s + 1).$$

For $G_1(s)$: time constant $T_1 = 1$ s, DC gain $K_1 = G_1(0) = 1$.

$$G_2(s) = 10/(s+10) = 1/(0.1*s + 1).$$

For $G_2(s)$: time constant $T_2 = 0.1$ s, DC gain $K_2 = G_2(0) = 1$.